

ESKDALEMUIR, United Kingdom

WMO: 031620

Lat: 55.312N

Lon: 3.207W

Elev: 242

StdP: 98.45

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.3	-4.4	-8.0	2.0	-4.1	-6.4	2.3	-2.6	13.6	6.3	12.3	6.6	0.9	340	0.661

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.3	22.8	16.7	20.6	15.6	18.8	14.8	17.6	21.2	16.5	19.4	15.6	17.8	3.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	11.9	18.7	15.3	11.2	17.4	14.6	10.6	16.5	50.5	21.3	47.0	19.5	44.3	17.8	21.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
11.2	9.8	8.6	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-9.2	26.0	2.3	1.9	-10.9	27.4	-12.3	28.5	-13.6	29.6	-15.3	31.0		
			-9.5	19.4	2.3	1.1	-11.2	20.2	-12.5	20.9	-13.8	21.5	-15.5	22.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.6	2.6	2.8	4.1	6.3	9.1	11.8	13.6	13.3	11.2	8.3	5.0	2.8
	DBStd	4.90	2.94	2.78	2.76	2.67	3.02	2.44	2.38	2.36	2.33	2.89	3.06	3.65
	HDD10.0	1249	228	201	184	114	54	9	1	2	13	67	151	224
	HDD18.3	3918	487	434	442	361	288	197	147	157	213	311	399	482
	CDD10.0	373	0	0	0	3	25	62	113	104	50	14	1	0
	CDD18.3	3	0	0	0	0	0	0	2	1	0	0	0	0
	CDH23.3	42	0	0	0	0	2	7	21	12	0	0	0	0
	CDH26.7	2	0	0	0	0	0	1	1	1	0	0	0	0
Wind	WSAvg	3.6	4.0	4.2	4.0	3.8	3.9	3.6	3.1	3.0	3.3	3.6	3.6	3.8
Precipitation	PrecAvg	1661	186	135	138	95	99	100	112	133	140	173	171	183
	PrecMax	2452	369	402	282	176	258	246	269	393	273	352	432	569
	PrecMin	1028	38	19	45	17	4	13	29	24	16	47	33	45
	PrecStd	277	87	81	59	39	54	44	50	77	59	84	77	96
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.1	10.6	14.4	18.9	22.8	24.1	25.8	25.2	20.8	16.2	13.1	11.1
		MCWB	9.1	8.6	8.9	12.1	15.7	16.6	18.3	18.3	15.8	13.3	12.4	10.4
	2%	DB	8.9	9.2	10.9	15.1	19.6	21.0	23.1	21.4	18.0	14.4	11.5	9.9
		MCWB	8.2	8.2	8.2	10.1	13.8	15.5	17.0	16.7	14.2	13.0	10.8	9.3
	5%	DB	7.9	8.1	9.4	12.6	16.9	18.7	20.6	19.0	16.5	13.5	10.7	9.0
		MCWB	7.2	7.3	7.6	9.0	12.2	14.3	16.0	15.6	13.7	12.3	9.9	8.5
	10%	DB	7.1	7.2	8.3	10.8	14.5	16.6	18.6	17.4	15.3	12.6	9.8	8.1
		MCWB	6.5	6.4	6.9	8.2	10.8	13.1	15.1	14.6	13.2	11.5	9.0	7.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.5	9.5	10.4	12.5	16.5	17.8	19.2	19.0	17.0	14.4	12.4	10.7
		MCDB	9.9	10.0	12.8	17.3	21.3	21.7	23.9	23.1	19.4	15.2	12.9	11.0
	2%	WB	8.3	8.4	9.0	10.7	14.3	16.1	17.8	17.4	15.5	13.5	11.1	9.5
		MCDB	8.7	8.9	10.3	13.6	18.5	20.0	21.7	20.3	16.7	14.1	11.4	9.8
	5%	WB	7.4	7.5	8.0	9.6	12.8	14.8	16.6	16.2	14.5	12.6	10.1	8.6
		MCDB	7.8	8.0	8.9	11.7	16.0	17.8	19.6	18.2	15.7	13.3	10.5	8.9
	10%	WB	6.6	6.5	7.2	8.7	11.5	13.7	15.5	15.3	13.6	11.7	9.1	7.5
		MCDB	7.1	7.1	8.2	10.4	13.9	15.9	17.9	16.8	14.9	12.5	9.7	8.0
Mean Daily Temperature Range	5% DB	MDBR	4.6	5.6	6.7	8.1	8.8	8.2	8.3	8.0	7.6	6.6	5.6	5.2
		MCDBR	5.2	6.2	8.5	11.6	13.2	12.4	12.3	10.6	9.7	6.7	5.7	5.5
		MCWBR	5.1	5.3	6.0	7.3	7.8	7.6	7.2	6.4	6.3	5.3	5.4	5.5
	5% WB	MCDBR	5.0	5.7	7.1	9.5	11.2	10.7	10.6	9.1	7.5	6.0	5.6	5.5
		MCWBR	5.2	5.2	5.6	6.5	7.0	7.0	6.6	6.0	5.6	5.3	5.5	5.6
Clear-Sky Solar Irradiance	taub	0.314	0.328	0.355	0.374	0.376	0.374	0.373	0.362	0.348	0.333	0.308	0.299	
	taud	2.252	2.347	2.305	2.282	2.321	2.352	2.354	2.399	2.444	2.432	2.334	2.225	
	Ebn at Noon	596	747	809	846	866	871	863	852	814	733	608	519	
	Edh at Noon	58	75	98	115	117	115	113	101	85	68	52	47	
All-Sky Solar Radiation	RadAvg	0.55	1.19	2.23	3.61	4.71	4.60	4.43	3.58	2.57	1.41	0.70	0.41	
	RadStd	0.08	0.12	0.23	0.39	0.36	0.45	0.49	0.28	0.21	0.17	0.07	0.04	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page